

GNSS Technologies and Receiver Testing

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GNSS Overview

Global Navigation Satellite Systems (GNSS) is the general term used for systems that allow users to determine their location based on information from satellites. In addition to the Global Positioning System (GPS) which is run by the United States, the GLObal Navigation Satellite System (GLONASS) from Russia, the Galileo system from the European Union, and the BeiDou Navigation Satellite System (BDS) from China, are also in operation.

Of the four main systems, GPS is the oldest GNSS, providing worldwide coverage and continues to be upgraded. The GPS system provided the Civil Initial Operational Capability (IOC) in 1993 and the Full Operational Capability (FOC) in 1995 with 24 satellites (Block II / IIA series) in full operation. New GPS III/IIIF satellites will be launched in future to enhance capabilities and provide additional signals and services.

The first GLONASS satellite was launched by the Soviet Union in 1982. The GLONASS system was formally declared operational in 1993 and reached a fully operational constellation of 24 GLONASS satellites in 1995. However, following the dissolution of the Soviet Union, the system experienced a decline. Russia restored and completed the system from 2002. The GLONASS system has been modernized and became fully operational consisting of “GLONASS-K” satellites in October 2011, providing worldwide coverage. Russia continues to enhance the GLONASS system with the launch of new GLONASS-K / K2 satellites which provide new signals similar to those of GPS and Galileo.

Galileo has been developed by the European Community and European Space Agency for many years. The first Galileo test satellite was launched in 2005. In 2011 and 2012, four in-orbit validation satellites as part of the operational system were launched. Galileo started offering Early Operational Capability (EOC) worldwide in December 2016 with 18 satellites in orbit. Federal Communications Commission (FCC) approved use of Galileo in the US in 2018. The Galileo system reached Full Operational Capability (FOC) in 2020 with 24 FOC satellites in full service. The system continues to be enhanced by launching new Galileo Second Generation (GSG) satellites.

The BeiDou Navigation Satellite System (BDS) is developed and operated by China in three phases. The first phase, BDS-1, was completed in 2000, providing services within China using three Geostationary Earth Orbit (GEO) satellites. The second phase, BDS-2, was completed in 2012, extending services to the Asia Pacific region with a constellation of 14 satellites (5 GEO, 4 MEO, 5 IGSO). The third phase, BDS-3, was completed in 2020, providing global services with a constellation that includes 24 MEO satellites, 3 MEO satellites and 3 IGSO satellites. The BDS system will optimize its mixed constellation architecture by developing and launching new-generation of satellites.

In addition to these four global satellite systems that allow receivers to determine their location, there are other related satellite systems to enhance position accuracy.

Satellite-Based Augmentation Systems (SBAS) use geostationary satellites to provide data to improve the positioning accuracy of GNSS receivers with specific regions. These include the Wide Area Augmentation System (WAAS) for North America, European Geostationary Navigation Overlay Service (EGNOS) for Europe, Multi-functional Satellite Augmentation System (MSAS) for Japan, GPS Aided Geo Augmented Navigation (GAGAN) system for India, System for Differential Corrections and Monitoring (SDCM) for Russia, BeiDou Satellite-Based Augmentation System (BDSBAS) for China and Southern Positioning Augmentation Network (SouthPAN) system for Australia and New Zealand.

Regional Navigation Satellite Systems (RNSS) provide services within specific regions and have the potential to evolve into global GNSS systems. These include Quasi-Zenith Satellite System (QZSS) for Japan and Asia-Oceania region using a 4-satellite constellation and the Indian Regional Navigation Satellite System (IRNSS, also known as NavIC) with 3 geostationary and 4 geosynchronous satellites.

Typically, the GNSS satellites have operational lifespans of 7 to 15 years. To maintain system integrity, new satellites are periodically launched to replace older ones. Enhancements have been implemented over the years and the development of advanced technologies and new signals will ensure that GNSS continues to meet the evolving demands of users worldwide.

This application note will focus on consumer devices and applications and provides a brief overview of GNSS system architecture, signal characteristics, and evolving applications, while highlighting the critical role of receiver testing in ensuring system reliability and performance.

GNSS system architecture

There are 3 major segments of the GNSS system.

- The space segment consists of the satellites or Space Vehicles (SV), which transmit a navigation message with information about the satellite's orbit and position and the time of the transmission. Each satellite is equipped with atomic clocks to ensure precise timing, which is critical for accurate positioning, navigation signal generators and solar arrays.
- The control segment consists of monitoring and control stations on the ground that track the satellites in view, collect data from the transmissions, and provide updates related to navigation information, atmospheric conditions, and corrections. They also manage satellite control functions, ensuring that satellites remain in their designated orbits and operate correctly.
- The user segment consists of devices equipped with GPS receivers that are capable of transforming signals from satellites into precise estimations of position, speed, and time.

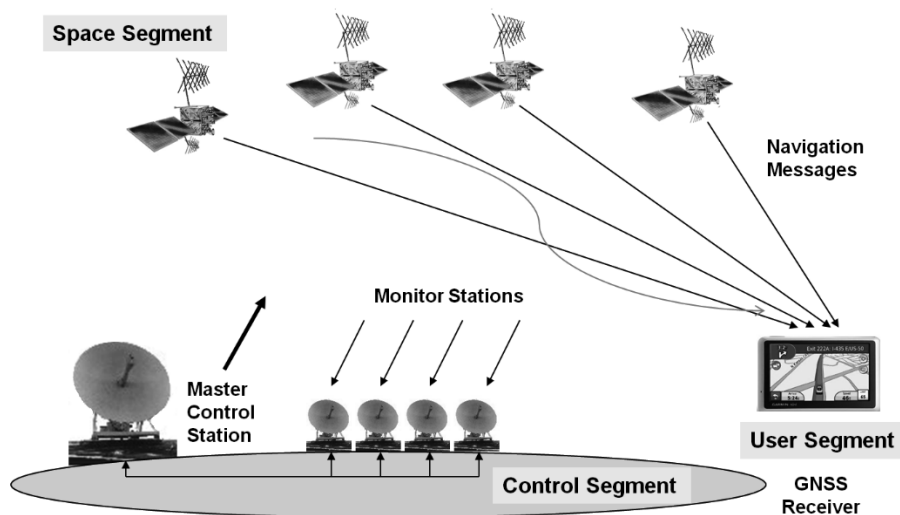


Figure 1. The three major segments of the GNSS system

GNSS receiver

From a high-level perspective, a GNSS receiver appears as an antenna which senses navigation signals, and output information such as status, positions, and time (via an I/O port or screen). As shown in Figure 2, a typical block diagram for GNSS receivers includes an antenna, an RF front end and down converter, a baseband processing element, and a computation engine. In some cases, these elements may be integrated in a single module which can be quite small – on the order of 1 square cm.

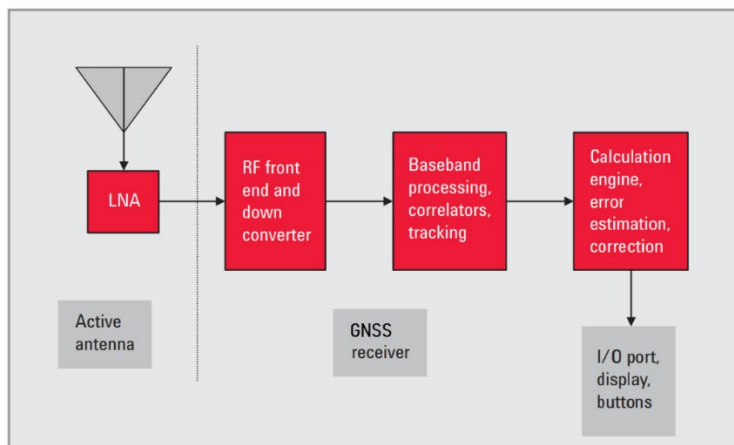


Figure 2. GNSS receiver block diagram

A GNSS receiver determines its location through a process called trilateration. It uses the transmitted time and position data in the navigation message to measure the time delay for the signal from the satellite and calculate its distance from that satellite, called the pseudorange. The distance from each satellite gives a sphere on which the receiver could be located. The intersection of two spheres is a circle, and the three spheres intersect at two points. A fourth data point is needed to identify the correct point as the receiver's location. For receivers that are assumed to be on the Earth's surface, the point that is on the Earth's surface would be the correct value. For a more general solution, the pseudorange for a fourth satellite is required. Simultaneous measurements from at least four satellites are also necessary to allow the receiver to correct clock errors and determine the correct time, in addition to determining its latitude, longitude, and altitude.

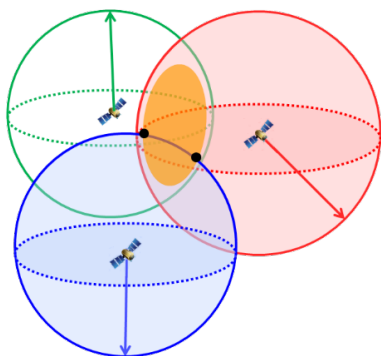


Figure 3. With trilateration, a GNSS receiver calculates the distance to at least 3 satellites. The intersections of the distance spheres around each satellite determine the location of the receiver.

GNSS satellites transmit relatively low power. At the Earth's surface, the power level is around -160 to -155 dBW (-130 to -125 dBm). Signal levels may be even lower where there are obstructions. In order to successfully recover the navigation signal, active antennas are typically utilized with gains of 20 or 30 dB. For these active antennas, a DC voltage is supplied from the receiver over the same cable that transmits the signal to the receiver. In most commercial implementations, the antennas are integrated into the receiver unit. In some cases, a port for an external antenna may be provided. Additionally, GNSS receivers include amplification and signal processing to allow them to recover the GNSS signals.

GNSS signal structure

The GNSS signal power is around -130 dBm on Earth's surface, requiring high-sensitivity receivers. The signal carrier frequencies are mainly within L-band (1–2 GHz) to balance atmospheric penetration and antenna size. The typical modulation schemes used by main GNSS signals include BPSK, BOC (Binary Offset Carrier), MBOC (Multiplexed BOC) for spectral efficiency and multipath mitigation, and QPSK.

Signal components

GNSS signals are structured around three fundamental components: **carrier**, **ranging codes**, and **navigation data**, which enable precise ranging, data transmission, and signal tracking. These components are optimized for robustness, interoperability, and performance in diverse environments.

Carrier

The carrier wave forms the backbone of signal transmission, modulated with ranging codes and navigation data. To improve spectrum efficiency, CDMA technique is used by most GNSS systems. Different GNSS systems share the same frequency carrier for interoperability.

When the satellite transmits more than one signal at the same time, the GNSS receiver can take advantage of multiple frequencies to improve its performance. For example, a dual-frequency GPS receiver takes use of a combined L2C + L1 C/A signal to enable ionospheric correction to greatly improve the accuracy to the military level (or better). Aircraft can use L5 in combination with L1 C/A to improve accuracy (via ionospheric correction) and robustness (via signal redundancy). When used in combination with L1 C/A and L2C, L5 will provide a highly robust service. Through a technique called trilateration, the use of three GPS frequencies may enable sub-meter accuracy without augmentations, and very long range operations with augmentations.

Ranging codes

Ranging codes are Pseudorandom Noise (PRN) sequences that are used for positioning and timing, signal spreading and precise delay estimation. Each GNSS system employs unique code designs to optimize performance, security, and interoperability.

There are different families of PRN codes. Usually, the ranging codes for public signals are defined in the Interface Control Document (ICD), providing standard precision service. The ranging codes for authorized signals, especially for military use, are encrypted and not released, providing high precision service.

For the GNSS system using CDMA technique, each satellite is assigned with a dedicated PRN code that is transmitted with navigation data. The high auto-correlation and low cross-correlation properties of these codes enable the receiver to correctly distinguish a signal and recover its original version by correlating the incoming signal with the desired PRN code.

The satellites in GPS, Galileo and BeiDou systems and the modernized GLONASS satellites that transmit CDMA signals utilize unique ranging codes to identify the satellite and enable accurate positioning.

Legacy GPS signals use C/A code for public signals providing standard precision service and P(Y) code for military use providing high precision service. Modernized GPS civilian signals use enhanced ranging codes with wider bandwidth and higher transmission power to minimize interferences and errors.

Galileo and BeiDou signals use long spreading codes generated from primary and secondary codes.

Table 1a. Ranging codes by system (CDMA-based)

GNSS	Ranging codes	Code type	Frequency	Features
GPS	C/A Code	Gold codes (1023 bits, 1.023 Mbps)	L1 (1575.42 MHz)	Open service, legacy standard for basic navigation
	P(Y) Code	Long-period PRN codes (encrypted) (6.1871x10 ¹² bits, 10.23 Mbps)	L1 (1575.42 MHz)	Military precision service, phased out by M code
			L2 (1227.60 MHz)	
	L1C	Weil codes (10230 chips, 1.023 Mcps)	L1 (1575.42 MHz)	Open service, modernized civilian signal, interoperable with Galileo E1 and BDS B1C
	M Code	Encrypted military codes	L1 (1575.42 MHz)	Military precision service, anti-jamming, secure, independent of civilian signals
			L2 (1227.60 MHz)	
	Civil Moderate (CM) Code	Gold code (10230 chips, 511.5 kcps)	L2 (1227.60 MHz)	Open service, higher effective power than the legacy L1 C/A signal, making it easier to receive under trees and even indoors
	Civil Long (CL) Code	Long-period code (767250 chips, 511.5 kcps)		
Galileo	I5-code	Neuman-Hofman (NH) code (10230 chips, 10.23 Mcps)	L5 (1176.45 MHz)	Open service, higher power, wide bandwidth to minimize multipath errors, advanced signal design
	Q5-code	Orthogonal modulated I5-code (10230 chips, 10.23 Mcps)		
	E1 OS	Gold code (data) + Memory code (pilot) (4092 chips, 1.023 Mcps) Secondary code: (25 chips)	E1 (1575.42 MHz)	Open service
	E5a Tiered code	Primary: Weil code (10230 chips, 10.23 Mcps) Secondary: fixed sequence (20 chips)	E5 (1176.45 MHz)	Open service for avionics navigation
GLONASS	E5b Tiered code	Primary: Gold code (10230 chips, 10.23 Mcps) Secondary: fixed sequence (4 chips)		For higher precision and availability
	E6 Tiered code	Encrypted codes	E6 (1278.75 MHz)	Commercial service and public regulated service (PRS)
	L1OC	Gold codes (data) (1023 chips, 1.023 Mcps) Kasami codes (pilot) (4092 chips,)	L1 (1600.995 MHz)	Open service
	L2OC	Gold codes (data) Kasami codes (pilot) (10230 chips,)	L2 (1248.06 MHz)	Open service
BeiDou	L3OC	Kasami codes (10230 chips, 1.023 Mcps)	L3 (1202.025 MHz)	Open service
	B1I OS	Gold codes (2046 chips, 2.046 Mcps)	B1 (1561.098 MHz)	Open service
	B1C Tiered code	Primary: Weil codes (10230 chips, 1.023 Mcps) Secondary codes: 1800 chips	B1C (1575.42 MHz)	Open service for higher precision and availability
	B2a Tiered code	Primary: long Weil codes (10230 chips, 10.23 Mcps) Secondary: fixed sequence	B2a (1176.45 MHz)	Open service
	B2b	Not published	B2b (1207.14 MHz)	Open service for precise point position (PPP)
	B3I	Gold codes	B3I (1268.52 MHz)	Open service

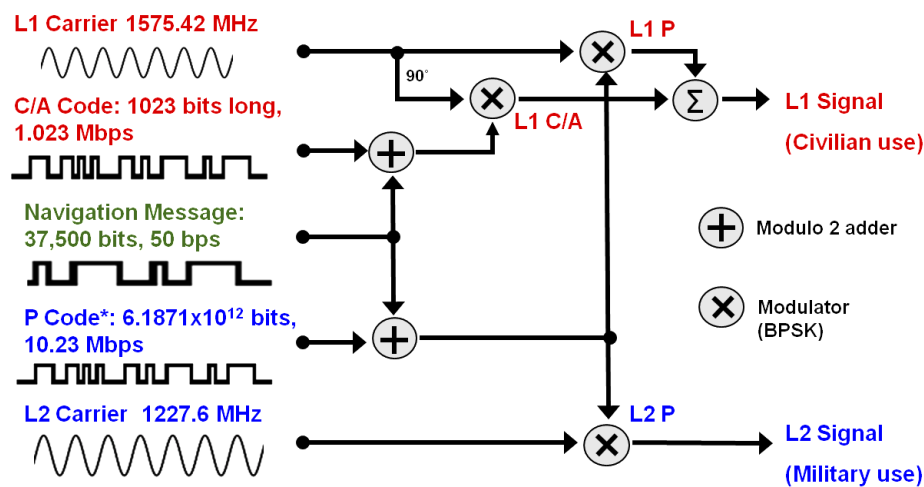
Legacy GLONASS satellites using FDMA technique broadcast both standard precision signals for public use and high precision signals for authorized users. The ranging codes consist of a ST code (similar to C/A code) for standard precision and a VT code (similar to P code) for high precision. The ST code is shorter than GPS C/A code, with a chip rate of 511 kbps, while the VT code has a chip rate of 5.11 Mbps. All satellites transmit the same ranging codes. GLONASS FDMA signals use a meander code combined with the ranging codes and navigation data, to improve bit boundary detection and enhance signal robustness.

Table 1b. GLONASS ranging codes (FDMA-based)

	Ranging codes	Code type	Frequency	Features
FDMA	ST code for standard precision (SP)	Gold codes (511 chips, 511 kcps)	L1 (1602.0-1615.5 MHz)	Civilian open service, frequency separation by satellite (spaced by 0.5625 MHz at L1, and by 0.4375 MHz at L2)
			L2 (1246.0-1256.5 MHz)	
	VT code for high precision (HP)	Long encrypted codes (33554432 chips, 5.11 Mcps)	L1 (1602.0-1615.5 MHz)	Enhanced anti-jamming via FDMA, for military use
			L2 (1246.0-1256.5 MHz)	

Navigation data

Navigation data provides satellite ephemeris, clock corrections, and system status, usually with low data rate. For example, GPS L1 C/A LNAV data stream transmitted at 50 bps data rate with subframe structure (e.g., almanac, ionospheric delay coefficients), Galileo E1-B data stream transmitted at 250 bps data rate with Forward Error Correction (FEC) and integrity flags, and BeiDou B1I data stream transmitted at 50 bps data rate with hybrid ephemeris parameters for regional/global services.



* GPS satellites transmit an encrypted version called P(Y) code

Figure 4. The example of legacy GPS signal shows a combination of the navigation message and the C/A or P code, modulated onto the L1 and L2 carriers.

Signal modulation

The following modulation schemes are adopted by the major GNSS systems:

- Binary phase shift keying (BPSK)
- Binary offset carrier (BOC)
- Multiplexed BOC (MBOC)
- Quadrature multiplexed BOC (QMBOC)
- Composite BOC (CBOC)
- Alternative BOC (AltBOC)
- Quadrature phase shift keying (QPSK)

BPSK features simple implementation and low power consumption, but it is susceptible to multipath interference and narrowband jamming due to concentrated spectral energy. BOC is in a better place than BPSK. The Galileo E1 OS signal uses CBOC modulation, transmitted at the same frequency as the GPS L1 signal. Using BPSK modulation, GPS L1 C/A spectrum shapes like a sinc(x) function with a main lobe at the center frequency. Using BOC modulation, Galileo E1 spectrum spreads the energy to two side lobes, leaving a null at the center frequency, which minimizes the interference with GPS.

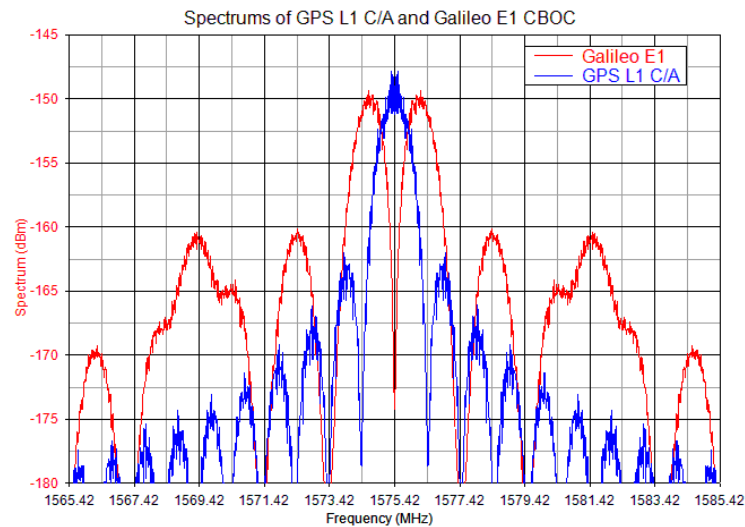


Figure 5. Spectrum of GPS L1 C/A signal compared to Galileo E1 CBOC signal.

BOC signals are commonly referred to as $\text{BOC}(m,n)$ or $\text{BOC}(f_s, f_c)$, where subcarrier frequency (f_s) = $m * 1.023$ MHz, and coding frequency (f_c) = $n * 1.023$ MHz. $\text{BOC}(m,n)$ splits spectrum into sidebands (e.g., $\text{BOC}(1,1)$ for Galileo E1), reducing interference with legacy signals, multipath resistance.

CBOC combines a narrowband $\text{BOC}(1,1)$ signal with a wideband $\text{BOC}(6,1)$ signal, with 50% power allocated to each, to improve multipath performance. MBOC combines $\text{BOC}(1,1)$ and $\text{BOC}(6,1)$ in a 4:1 power ratio, minimizing inter-system interference. AltBOC supports wide bandwidth with high precision.

QPSK modulation is introduced for high data rate with interference resilience by advanced satellites, such as BDS-3 satellites and GLONASS L3 satellites.

Table 2. Modulation schemes and multiple access methods by system

GNSS	Signal	Frequency	Modulation	Multiple-Access
GPS	L1 C/A	1575.42 MHz	BPSK	CDMA
	L1C	1575.42 MHz	MBOC	CDMA
	L2C	1227.60 MHz	QPSK	CDMA
	L5	1176.45 MHz	BPSK	CDMA
Galileo	E1 OS	1575.42 MHz	CBOC	CDMA
	E5a	1176.45 MHz	AltBOC	CDMA
	E5b	1207.14 MHz	AltBOC	CDMA
	E6	1278.75 MHz	BPSK	CDMA
GLONASS	L1OF	1602.0-1615.5 MHz	BPSK	FDMA
	L1OC	1600.995 MHz	BPSK (data), BOC (pilot)	CDMA
	L2OF	1246.0-1256.5 MHz	BPSK	FDMA
	L2OC	1248.06 MHz	QPSK	CDMA
	L3OC	1202.025 MHz	QPSK	CDMA
BeiDou	B1I OS	1561.098 MHz	QPSK	CDMA
	B1C	1575.42 MHz	MBOC	CDMA
	B2a	1176.45 MHz	QPSK	CDMA
	B2b	1207.14 MHz	QPSK	CDMA
	B3I	1268.52 MHz	BPSK	CDMA

Signal multiple access methods

Two multiple access methods are used by satellite signal transmission: Frequency-Division Multiple Access (FDMA) and Code Division Multiple Access (CDMA). In FDMA, all satellites transmit the same PRN code but at different carrier frequencies, whereas in CDMA all satellites transmit at the same carrier frequency but with a different PRN code, that is assigned to each satellite.

GLONASS uses FDMA for its legacy signal transmission. Originally, the system transmitted the signals within two bands: L1, 1602.0–1615.5 MHz, and L2, 1246.0–1256.5 MHz, at frequencies spaced by 0.5625 MHz at L1 and by 0.4375 MHz at L2, which provided 25 channels, so that each satellite in the full 24-satellite constellation could be assigned a unique frequency. The system now uses only 14 primary frequency channels for 24 satellites because frequency channels are reused for satellites on opposite sides of the Earth.

The CDMA technique is used by GPS, Galileo, and BeiDou satellites. The satellites that transmit signals on the same frequency band are distinguished by unique codes (ranging codes). The modernized GLONASS satellites also transmit CDMA signals which are expected to notably improve and facilitate precise point positioning applications once fully deployed across the GLONASS constellation.

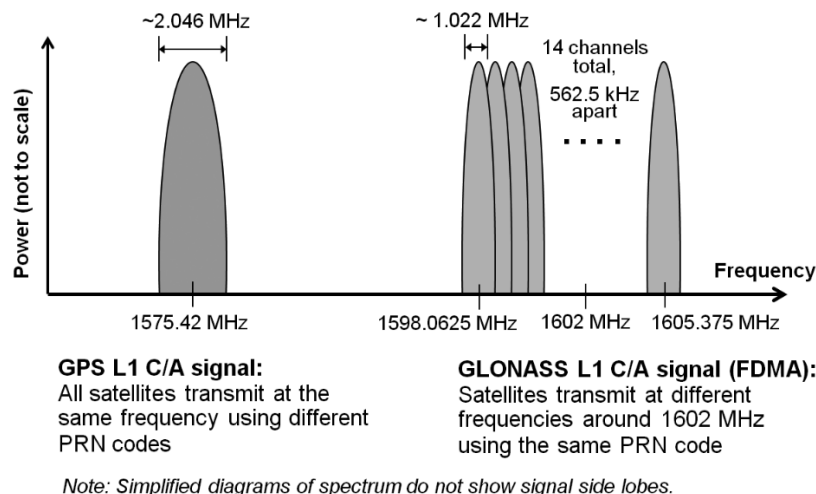


Figure 6. Comparison of the CDMA signal for GPS and the FDMA signals for GLONASS

GNSS technology applications and growth drivers

The growth of the GNSS market is being driven by several factors, primarily the increasing demand for location-based services and applications integrated into various consumer devices, for example, smart phones, tablets, digital cameras, smartwatches, and fitness trackers.

Another factor is the increasing demand for fleet management and logistics applications. Companies are increasingly utilizing GNSS technology to track vehicles in real-time, optimizing routes and improving operational efficiency. Personal tracking applications are also gaining traction, providing safety solutions for monitoring vulnerable populations, such as the elderly or children. Law enforcement agencies use GNSS technology to manage tracking of individuals under house arrest or parole, enhancing public safety measures. Additionally, the market for pet tracking devices has expanded, offering solutions for pet owners to keep tabs on their animals.

In automotive industry, the development of autonomous driving drives strong demand for high precision positioning – now every car is equipped with a GNSS receiver for location-based applications.

To support this growth, manufacturers of WLAN and cellular chipsets are actively developing GNSS chipsets and modules. The demand for devices that can support multiple GNSS constellations has led to the emergence of multi-GNSS chipsets, enabling compatibility with GPS, GLONASS, Galileo, and BeiDou, among others. This multi-constellation capability enhances positioning accuracy and reliability, especially in challenging environments.

Beyond consumer applications, GNSS technology is also crucial in high-precision commercial markets such as surveying, agriculture, and aviation.

This application note will focus on consumer devices and applications.

GNSS Receiver Testing

For most wireless digital systems, receiver testing involves testing Bit Error Rate (BER), Frame Error Rate (FER), Packet Error Rate (PER), or Block Error Rate (BLER) under specific power, noise, fading and interference conditions. As GNSS navigation message data is transmitted at only 50 bits per second (bps), achieving a typical BER of 1 error in 10^6 would require an extensive testing time, so BER testing is impractical for GNSS receivers. Therefore, GNSS receiver testing focuses on the receiver's ability to acquire and track satellite signals and calculate location fixes under varying conditions.

The following sections outline the key test requirements and methodologies for GNSS receiver testing.

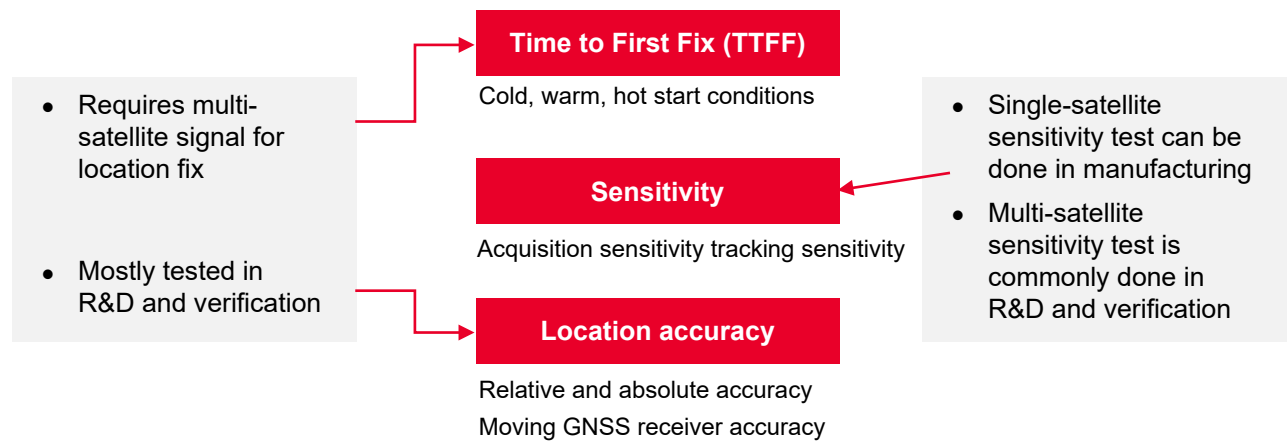


Figure 7. Typical GNSS receiver tests

The most common GNSS receiver verification tests include Time to First Fix (TTFF), receiver sensitivity, and location accuracy. Other tests are also performed, such as re-acquisition time, interference testing, antenna performance, 1 Pulse per Second (PPS) timing accuracy, special events, and robustness, although less common.

Time to first fix (TTFF)

When a receiver is turned on, it must do some searching to find the satellite signals – this process is called acquisition. It must then track the signals and compute a position. The time from power-on to the availability of the first valid location fix is called TTFF. TTFF is a critical performance metric for GNSS receivers, directly influencing user satisfaction by minimizing the time required to obtain a location fix.

TTFF testing can be categorized into cold, warm, and hot start conditions. These terms refer to the data that is available to the receiver when it is turned on. GNSS receivers can be put into these states using receiver control software provided by the chipset vendor.

Cold start TTFF

In a cold start, the receiver has no data about the satellites in its memory, so it must search across all possible PRN codes and a frequency range of approximately ± 5 kHz of Doppler shift. For GPS, the receiver must receive at least 18 seconds of good data from each satellite in order to receive an accurate description of the satellite's orbit (known as ephemeris data) for each satellite to determine the satellite's location. The ability of modern receivers to process signals in parallel and utilize advanced techniques allows them to achieve cold start TTFF in less than a minute, with typical values ranging from 30 to 45 seconds.

During testing, the receiver is placed in a cold start state usually by sending a command through a test connection. A strong satellite signal is then transmitted to the receiver. The time from power-on to the establishment of the first valid location fix is recorded as the cold start TTFF.

For design verification testing, multiple cold start TTFF tests are conducted across various conditions, such as different locations, times, and satellite geometries, to cover a wide range of operational scenarios. The results are averaged to provide a robust characterization of the receiver's performance under diverse conditions. The test purpose is to ensure that the receiver design can reliably perform across different real-world situations. Consistency is maintained by performing the tests within similar environmental setups, although variations in satellite visibility and positioning may still influence the results.

In contrast, manufacturing testing typically selects a controllable environment. In this scenario, several cold start TTFF tests are conducted under identical conditions, such as the same time of day and satellite visibility. The purpose is to test the receiver's performance consistency across multiple units, leading to averaged results that reflect manufacturing variations. By utilizing a standard testing protocol, manufacturers can ensure that all produced units meet the specified performance criteria.

Warm start TTFF

In a warm start, the receiver retains the approximate time, last location, and almanac data. This retained information allows the receiver to significantly reduce the time required to achieve a location fix compared to a cold start.

Warm start TTFF testing is less common than cold start TTFF testing due to its specific requirements, that is, the receiver must first be exposed to the satellite signals for approximately 15 minutes to ensure it has received the complete almanac data and then perform the TTFF test. The test typically begins by sending a "warm start" command to the receiver. The time from the command sent to the establishment of the first valid location fix is recorded as the warm start TTFF.

However, repeating the warm start TTFF test under consistent conditions can be challenging because it requires maintaining the same environmental factors and ensuring that the receiver is in the appropriate state, specifically having the necessary almanac data, at the time of testing. For instance, conducting the test exactly 15 minutes into a scenario can be time-consuming and it requires a signal generator that can be programmed to start the test precisely at that 15-minute interval.

Modern GNSS receivers can achieve a warm start TTFF in as little as 1 to 5 seconds, leveraging the retained data to quickly lock onto available satellites. This rapid response time enhances the usability of

GNSS technology in applications where timely location fixes are critical, such as in automotive systems, mobile devices, and other real-time navigation applications.

Hot start TTFF

In a hot start, the receiver also remembers the last satellites that were used and tries to rapidly acquire them again. This may work if the receiver has only been off for a short time (typically less than one to two hours) and has not moved significantly (within approximately 100 meters).

Hot start testing is less common than cold start TTFF testing, but it is more frequent than warm start TTFF testing. The test begins by sending a “hot start” command to the receiver. Unlike warm start TTFF, a hot start does not require the receiver to be exposed to satellite signals for an extended period (such as 15 minutes). Instead, the receiver can quickly reacquire satellites based on the retained information, often within minutes.

Modern GNSS receivers can achieve hot start TTFF in remarkably short times, typically around 1 to 2 seconds, depending on the specific technology and conditions. This efficiency is particularly valuable when requiring immediate location fixes, such as automotive navigation, drone navigation, mobile devices and emergency services.

Sensitivity

Sensitivity refers to the minimum level of signal required for a receiver to acquire or track a satellite. GNSS receivers have two different sensitivity levels: acquisition sensitivity and tracking sensitivity.

Acquisition sensitivity

Acquisition sensitivity refers to the minimum signal level required for a receiver to successfully perform a cold start TTFF within a specified timeframe. During the acquisition process, the required signal level must be higher than during the tracking process, because the time synchronization is not established. The required signal level is defined as the minimum power level that allows for a cold start TTFF of 100 seconds or less.

One type of acquisition sensitivity test is the single-satellite sensitivity test. In this test, a signal with a known amplitude and a static Doppler shift is presented to the receiver. A receiver cold start is performed, and the time to acquisition is recorded. The signal power levels are then gradually reduced until the receiver can no longer acquire the signal. This test can be repeated at different Doppler shifts to create a set of curves depicting the power versus acquisition time across various Doppler conditions. These curves can be used to characterize the acquisition sensitivity of a receiver, helping to understand its performance in diverse operational scenarios.

Tracking sensitivity

Tracking sensitivity refers to the minimum signal level required for the receiver to maintain a location fix within a specified degree of accuracy. This level is generally much lower than the acquisition sensitivity level. As the signal level is reduced, the receiver’s ability to recover the navigation message data stream decreases, resulting in potential bit errors. However, because the Doppler frequency and the timing of the signal are known, the tracking loops can still operate successfully.

As the signal levels continue to decrease, the noise within the tracking loops becomes relatively higher, which can degrade the timing and/or frequency synchronization. Consequently, the accuracy of the location fix is adversely affected. Eventually, as the signal level continues to decrease, the system will incrementally lose its ability to track satellites, until the receiver is not able to compute a location fix.

Typically, tracking sensitivity is measured as the minimum power required to maintain a specific level of location accuracy. This measurement is dependent on the scenario and the time into the scenario, so it should be averaged over multiple tests conducted at different times in different scenarios.

Location accuracy

Location accuracy refers to the receiver's ability to achieve a location fix that is as close as possible to the desired position, both in terms of repeatability and accuracy. Variations of location accuracy testing include:

- **Relative location accuracy test:** This test compares the location fixes obtained by cold, warm, and hot starts at the same location and analyzes the variation between fixes. A low variation indicates a reliable location fix, beneficial for returning to the same location
- **Absolute location accuracy test:** In this test, the location fixes obtained by cold, warm and hot starts at the same location are compared to the ideal location provided by the scenario. The deviation from the intended position is measured to assess the accuracy of the receiver's location fix
- **Dynamic location accuracy tests:** In addition to location accuracy, the accuracy of velocity is also considered as an important metric for a moving receiver. These tests simulate receiver movement while conducting accuracy evaluation to assess performance under dynamic conditions. This type of tests helps to understand how well the receiver maintains accuracy while in motion

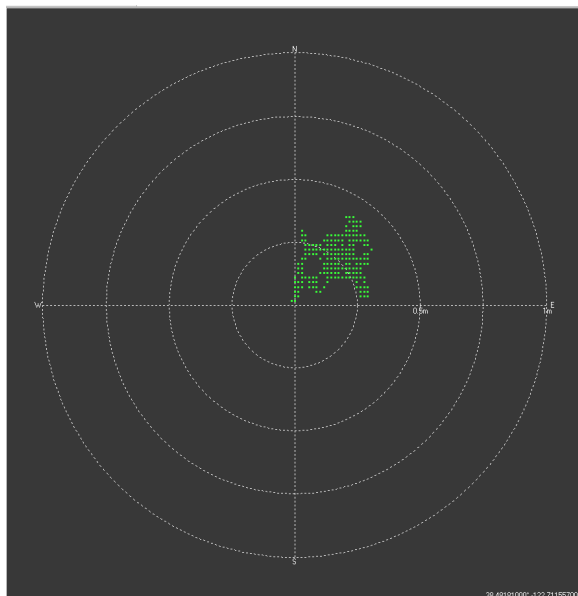


Figure 8. GNSS location accuracy – measured location vs. ideal location

Reacquisition time

Reacquisition time testing evaluates the receiver's performance when satellite signals are significantly reduced or temporarily interrupted, followed by restoration. This scenario might occur when a vehicle is going through a tunnel or moving under heavy tree cover. In this case, the receiver may briefly lose the ability to track most or all satellites but must re-acquire the signals once visibility is restored. To simulate this scenario, the signal generator power can be briefly reduced or turned off and then restored without restarting the testing scenario. Additionally, a related test involves interrupting signals from only a subset of satellites — for instance, when driving behind a building or hill that temporarily blocks signals from a portion of the sky.

The results from this test are typically compared against signals above the minimum sensitivity levels, representing good signal conditions. This comparison helps evaluate the receiver's ability to quickly re-acquire signals under less favorable conditions, providing insights into its robustness and reliability in real-world applications.

Antenna testing

In car navigation tests, antenna patterns can be used to emulate the blocking of satellite signals from the car body. In the real world, no antenna is ideal; therefore, real antennas do not present an isotropic response pattern. This means that the same signal arriving at the antenna from different points in the sky may produce varying signal strengths and phases at the receiver front end.

To address this variability, the test GNSS signal must replicate these conditions during conductive testing by allowing users to input specific antenna response patterns. The simulator then adjusts the signal strength from satellites accordingly. This approach represents a meta-test, where the preceding tests are repeated using different antenna patterns to evaluate how variations in antenna performance impact overall receiver functionality.

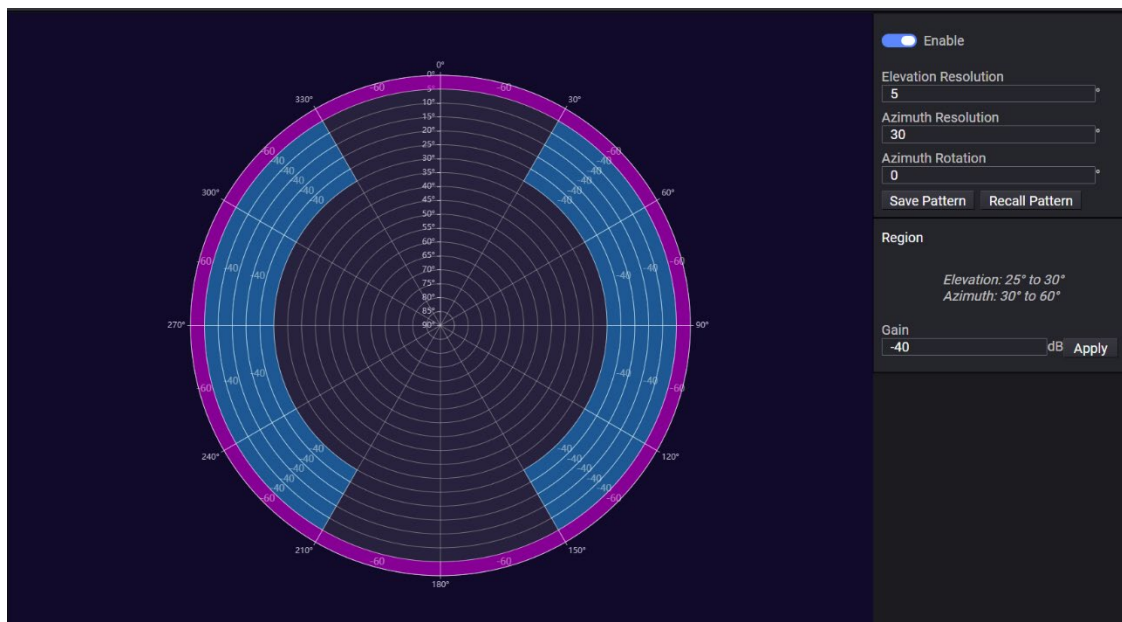


Figure 9. Antenna pattern testing

1 PPS timing accuracy

After acquiring signals from multiple satellites, the GNSS receiver can derive 1 PPS (One Pulse Per Second) output that is well aligned with the second of GPS time. Some models of GNSS receivers feature a dedicated pin or connector for outputting the 1 PPS signal for external timing purpose. 1 PPS timing accuracy may vary significantly, ranging from the nanosecond to microsecond level, depending on the receiver models and the specific test conditions. Therefore, it is critical to verify that the timing accuracy level can meet the design specifications.

The typical method for testing 1 PPS timing accuracy involves using a GNSS simulator to output satellite RF signals to the receiver, together with the aligned 1 PPS pulse. The test steps include:

1. Connect the output of the GNSS simulator, which provides the ideal 1PPS signal, to an oscilloscope.
2. Connect the 1 PPS output from the GNSS receiver under test to the same oscilloscope.
3. Use the ideal 1 PPS signal from the GNSS simulator as the trigger source on the oscilloscope.
4. Compare and collect the Root Mean Square (RMS) data of the time gap between the two 1 PPS pulses over a certain time interval.

The following example illustrates the 1 PPS timing accuracy measurement: the yellow line is 1 PPS from the GNSS simulator and the green lines indicate the persistence for receiver's historical 1 PPS pulses.

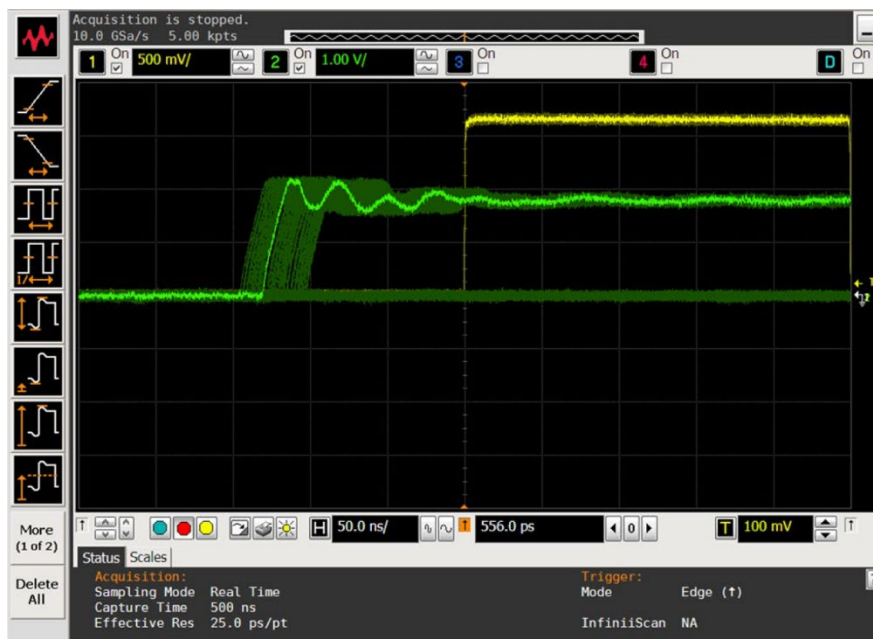


Figure 10. 1 PPS timing accuracy measurement on the oscilloscope

Special events

There are two special events in the GNSS systems that occur at regular intervals of several years. Any GNSS receiver must be able to behave correctly during these events to ensure its normal operation. Typically, these tests are conducted with a GNSS simulator capable of creating the required scenarios for these special events.

GPS rollover

The system time in navigation message is coded as a combination of Week Number (WN) and Time of Week (TOW) with a fixed number of bits. For example, GPS uses a 10-bit week number, which represents a rollover period of 1024 weeks, approximately 19.6 years. As of today, the next GPS rollover is expected to occur at midnight on November 20th to 21st, 2038. To verify that a receiver can pass smoothly through this rollover point, a GNSS simulator must run a scenario that spans this critical time, ensuring that the UTC reported by the receiver remains continuous.

Leap second

A leap second is an occasional one-second adjustment made to UTC time to keep it synchronized with the Earth's rotation. In contrast, GPS time operates continuously without leap seconds applied, resulting in a constant offset between GPS time and UTC time. As of 2025, this offset is +18 seconds. To verify that a GNSS receiver can properly handle a newly added leap second, a GNSS simulator must run a scenario where a leap second occurs at the scheduled time (either at the end of June 30th or December 31st). The test will check that the UTC time reported by the receiver displays the leap second accurately as 23:59:60.

Noise and interference

In today's environment, GNSS receivers are often used in complex electromagnetic conditions, which include white noise, cellular and wireless networks operating on the adjacent bands, and various intentional or unintentional interferences that can affect GNSS signals. To evaluate the anti-jamming performance of a GNSS receiver in a particular environment, an effective method is to use a GNSS simulator capable of outputting GNSS satellites signal along with configurable noise or interference signals. Some typical test scenarios include:

- Additive White Gaussian Noise (AWGN): Configure AWGN noise at the required Signal-to-Noise Ratio (SNR) level, then monitor the drop in carrier-to-noise ratio (C/N) reported by the receiver against a defined threshold (for example, 1dB)
- In-band and out-of-band interference: Simulate occasional in-band or out-of-band interference using Continuous Wave (CW) or chirp signals. Evaluate the receiver's performance degradation in terms of TTFF, sensitivity, and location accuracy
- Co-existence with other wireless signals: Simulate an environment where GNSS signals coexist with other wireless signals, including Bluetooth, LTE, Wi-Fi, and 5G. Assess the receiver's performance degradation in terms of TTFF, sensitivity and location accuracy
- Multipath conditions: Simulate signals that reflect off obstacles (for example, buildings or mountains) resulting in multipath propagation. This scenario can lead to issues such as degraded location accuracy, increased TTFF, or longer reacquisition time

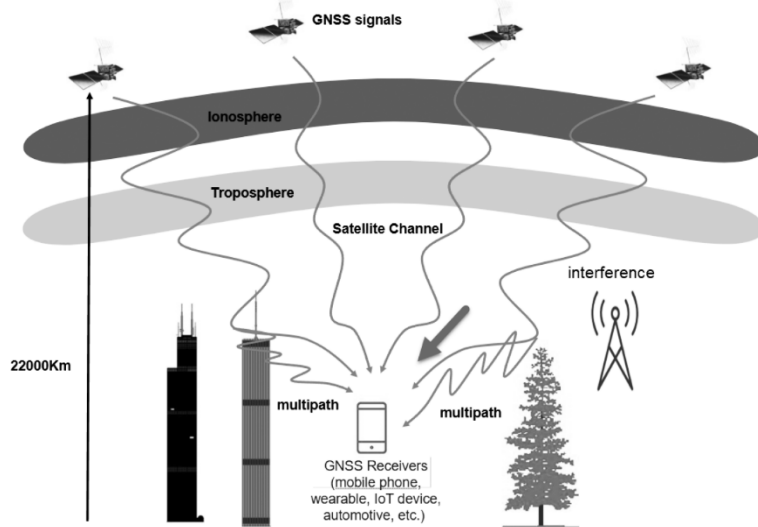


Figure 11. GNSS signal propagation involves free-space loss, ionospheric delay, and multipath reflections

Robustness

The robustness test ensures that a GNSS receiver can maintain its availability and reliability under exceptional circumstances, such as satellite health issues. To create such test scenario, a GNSS simulator must have capabilities like user-defined navigation messages and real-time control over satellite status. Potential exceptional scenarios include:

- Heal flag modification: Change health flag of one or more simulated satellites to off in the navigation message, verify that those satellites are no longer used by GNSS receiver by checking its NMEA messages output
- Satellite clock error: Add a fixed or drifting clock error (e.g. 100ns) in the simulated signals for some satellites from one constellation, then check the location deviation does not exceed a certain threshold
- Pseudo-range error: Apply user-defined pseudo-range errors (for example, 2 meters) to one or more simulated satellites. Confirm that the location accuracy still meets the operational requirements

Atmosphere modeling

Atmospheric conditions in the ionosphere and troposphere can introduce additional errors in time of arrival and signal strength, typically leading to degraded location accuracy. These atmospheric effects can include:

- Ionospheric delays: Variations in the ionosphere can cause delays in the signal, affecting the accuracy of the positioning data
- Tropospheric delays: The troposphere can also cause delays, particularly under varying weather conditions, which can further impact the receiver's accuracy

To account for these errors, advanced GNSS simulators can model atmospheric conditions, allowing for comprehensive testing of receiver performance under realistic scenarios.

GNSS signal simulator requirements

It is possible to test a GNSS receiver by using an antenna to capture off-the-air signals. While this approach is realistic, it can only provide limited information because the signals presented to the receiver are highly variable and non-repeatable. There may not be adequate satellites signals available at a particular location and time. In addition, testing under specific conditions, such as in particular locations or at high velocities, can become expensive and impractical.

A record-and-playback system can provide simulated GNSS signals to test the receiver. Although this system does provide repeatable test signals, it lacks the flexibility to modify the recorded signals, adjust individual satellite signals, or add impairments in real-time while the signals are playing back.

To overcome these challenges, a real-time GNSS signal simulator is recommended. A GNSS signal simulator generates output signals that accurately model what a GNSS receiver would receive: a mix of signals from many different satellites at different time delays, Doppler shifts, and power levels. The simulator can also replicate signal properties, including path losses, atmospheric distortions, relativistic effects, and transit time variations as signals travel from the satellite transmitter to the receiver antenna.

The real-time capabilities of a GNSS simulator allows modifications to the signal while it is being generated. Another advantage is that a GNSS simulator can simulate satellites that are not yet operational, facilitating early-stage testing and validation of systems prior to full satellites deployment.

The following key features are essential for a GNSS simulator to effectively perform comprehensive GNSS receiver testing.

Table 3. GNSS signal simulation requirements for GNSS receiver testing and validation

Feature	Test requirements or challenges addressed
Multiple satellites simulation	Simulate a minimum of four satellites per constellation to validate TTFF and ensure adequate satellite visibility
Dual-band signal support	Test TTFF, position accuracy, and the receiver's ability to utilize multiple frequencies for location calculations
Multiple GNSS constellations (GPS, GLONASS, Galileo, BeiDou)	Simulate GPS, GLONASS, Galileo, and BeiDou to assess position accuracy and the receiver's capability to combine signals from various constellations
Stationary and moving receiver scenarios	Evaluate TTFF and tracking performance in dynamic environments, ensuring the receiver can maintain a location fix while in motion
Custom scenario generation	Enable testing under specific conditions for any date, time, and location, replicating real-world scenarios encountered by the receiver
Real-time control of satellite visibility and power	Assess receiver sensitivity and its ability to reacquire signals under diverse operational conditions
Multipath signal support	Simulate reflected signals from obstructions to evaluate real-world performance
Elevation mask configuration	Model visibility loss due to obstructions like tunnels or mountains by restricting satellites to those above a specified elevation angle
Atmospheric modeling	Incorporate ionospheric and tropospheric effects, addressing atmospheric attenuation and delay for realistic signal simulation
Noise control	Add calibrated additive white Gaussian noise (AWGN) to precisely control the carrier-to-noise (C/N) ratio
Antenna modeling	Simulate the effects of different antenna types for realistic receiver testing
1 PPS output signal	Test 1 PPS timing accuracy, essential for applications requiring precise timing
User friendly interface	Provide an intuitive and user-friendly interface for easy setup and operation
Remote command control	Facilitate automation and efficiency in testing procedures

Keysight solution for GNSS receiver testing

N7609APPC PathWave Signal Generation for GNSS embedded application, running on the M9484C VXG vector signal generator, offers a real-time satellite simulation solution capable of generating multiple constellations and multiple frequencies signals for GNSS receiver testing. Using user-defined scenarios for stationary receivers or moving receivers, the application takes real-time control of the visibility and power of individual satellite, supports various models for troposphere and ionosphere, and provides various real-time status displays, including sky view, power view and trajectory view.



Figure 12. PathWave Signal Generation for GNSS embedded application running on the M9484C VXG signal generator

Conclusion

The rapid growth of location-based services and applications, increasing integration of navigation receivers into consumer devices, and the expansion of GNSS systems beyond GPS, with deployment of newer systems such as GLONASS, Galileo, and BeiDou, present new challenges for testing GNSS receivers.

To ensure thorough testing of GNSS receivers, it is essential to have the capability to simulate multiple GNSS constellations and multi-frequency bands. This includes the ability to accurately model various impairments in satellite signals, such as multipath effects, atmospheric distortions, and interference scenarios. Furthermore, real-time GNSS signal simulation is critical, allowing for dynamic adjustments to satellite visibility, power levels, and other conditions that the receiver may encounter in real-world applications.

A state-of-the-art GNSS simulator equipped with these features will enable manufacturers and developers to rigorously test their receivers, ensuring optimal performance in diverse environments. By addressing these requirements, the industry can enhance the reliability and accuracy of GNSS solutions, ultimately supporting the growing demands of modern navigation and positioning applications.

Websites

Official GPS website: gps.gov

Official GLONASS website (English version): glonass-iac.ru/en

Official Galileo website (English version): gsc-europa.eu

Official BeiDou website (English version): en.beidou.gov.cn

Keysight PathWave Signal Generation for GNSS: www.keysight.com/find/N7609APPC

Related Literature

Keysight N7609C Signal Studio for Global Navigation Satellite Systems (GNSS) Technical Overview, Literature number [5992-2740EN](#)

PathWave Signal Generation for GNSS Embedded Application Technical Overview, Literature number [3125-1001EN](#)

GPS Receiver Testing, Application Note, Literature number [5990-4943EN](#)



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